CMPG 322 - Decision Support Systems (DSS) 2

SU7 - Simulation Models: Introduction, Advantages and Disadvantages



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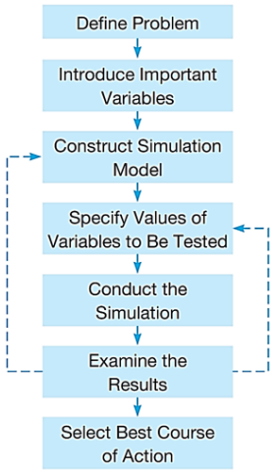
Outline

- Explain the advantages and disadvantages of simulation;
- Understand the five steps of conducting a Monte Carlo simulation;
- Analyse a simulation model as applied to inventory control;
- Analyse a simulation model as applied to queuing theory;
- Analyse a simulation model as applied to machine maintenance; and
- Understand the other two types of simulation models: operational gaming and systems simulation.

- Simulation is one of the most widely used quantitative analysis tools
- To simulate is to try to duplicate the features, appearance, and characteristics of a real system
  - Build a mathematical model that comes as close as possible to representing the reality of the system
  - Physical models can also be built to test systems
  - Problems can range from simple to extremely complex

Using simulation, a manager should:

1. Define a problem
2. Introduce the variables associated with the problem
3. Construct a simulation model
4. Set up possible courses of action for testing
5. Run the simulation experiment
6. Consider the results
7. Decide what courses of action to take



Advantages of Simulation

1. Relatively straightforward and flexible
2. Recent advances in computer software make simulation models very easy to develop
3. Can be used to analyse large and complex real-world situations
4. Allows "what-if?" type questions
5. Does not interfere with the real-world system
6. Enables study of interactions between components
7. Enables time compression
8. Enables the inclusion of real-world complications

Disadvantages of Simulation

1. Often expensive, may require a long complicated process to develop the model
2. Does not generate optimal solutions; it is a trial-and-error approach
3. Requires managers to generate all conditions and constraints of real-world problem
4. Each model is unique and the solutions and inferences are not usually transferable to other problems

- When systems contain elements that exhibit chance in their behavior, the Monte Carlo method of simulation can be applied
- The basis of the Monte Carlo simulation is experimentation on the probabilistic elements through random sampling
  - Some examples are:
    1. Inventory demand
    2. Lead time for inventory
    3. Times between machine breakdowns
    4. Times between arrivals
    5. Service times
    6. Times to complete project activities
    7. Number of employees absent

Based on these five steps:

1. Establishing a probability distribution for important input variables
2. Building a cumulative probability distribution for each variable in Step 1
3. Establishing an interval of random numbers for each variable
4. Generating random numbers
5. Simulating a series of trials

Example: Harry's Auto Tyre

A popular radial tire accounts for a large portion of the sales.

- Determine a policy for managing this inventory
- Simulate the daily demand for a number of days

Step 1: Establishing probability distributions

- One way to establish a probability distribution for a given variable is to examine historical outcomes
- Managerial estimates based on judgment and experience can also be used

TABLE 13.1 Historical Daily Demand for Radial Tires at Harry's Auto Tire and Probability Distribution

Demand for Tires	**Frequency (Days)**	**Probability of Occurrence**
0	10	$10 \div 200 = 0.05$
1	20	$20 \div 200 = 0.10$
2	40	$40 \div 200 = 0.20$
3	60	$60 \div 200 = 0.30$
4	40	$40 \div 200 = 0.20$
5	30	$30 \div 200 = 0.15$
Total	200	$200 \div 200 = 1.00$

- **Step 2:** Building a cumulative probability distribution for each variable
  - Converting from a regular probability to a cumulative distribution is an easy job
  - A cumulative probability is the probability that a variable will be less than or equal to a particular value
  - A cumulative distribution lists all of the possible values and the probabilities

TABLE 13.2 Cumulative Probabilities for Radial Tires

DAILY DEMAND	PROBABILITY	CUMULATIVE PROBABILITY
0	0.05	0.05
1	0.10	0.15
2	0.20	0.35
3	0.30	0.65
4	0.20	0.85
5	0.15	1.00

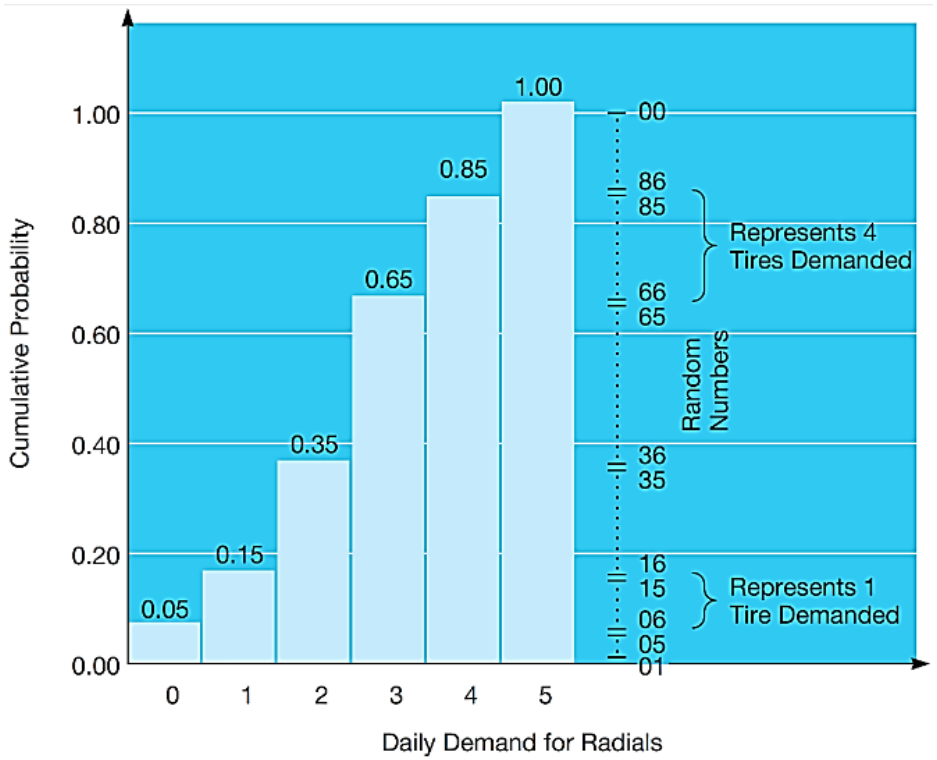
- **Step 3:** Setting random number intervals
  - *Assign a set of numbers to represent each possible value or outcome*
  - *These are* ***random number intervals***
  - *A* ***random number*** *is a series of digits that have been selected by a totally random process*
  - *The range of the random number intervals corresponds exactly to the probability of the outcomes*
  - *A cumulative probability graph can help assign random numbers*

TABLE 13.3 Assignment of Random Number Intervals for Harry's Auto Tire

DAILY DEMAND	PROBABILITY	CUMULATIVE PROBABILITY	INTERVAL OF RANDOM NUMBERS
0	0.05	0.05	01 to 05
1	0.10	0.15	06 to 15
2	0.20	0.35	16 to 35
3	0.30	0.65	36 to 65
4	0.20	0.85	66 to 85
5	0.15	1.00	86 to 00

Example: Harry's Auto Tyre

FIGURE 13.2 Graphical Representation of the Cumulative Probability Distribution for Radial Tires



- **Step 4:**
  - Generating random numbers
  - Random numbers can be generated in several ways
  - Large problems will use computer program to generate the needed random numbers
  - For small problems, random processes like roulette wheels or pulling chips from a hat may be used
  - The most common manual method is to use a random number table
  - Everything is random in a random number table so numbers can be selected from anywhere in the table

TABLE 13.4 Table of Random Numbers

52	06	50	88	53	30	10	47	99	37
37	63	28	02	74	35	24	03	29	60
82	57	68	28	05	94	03	11	27	79
69	02	36	49	71	99	32	10	75	21
98	94	90	36	06	78	23	67	89	85
96	52	62	87	49	56	59	23	78	71
33	69	27	21	11	60	95	89	68	48
50	33	50	95	13	44	34	62	64	39
88	32	18	50	62	57	34	56	62	31
90	30	36	24	69	82	51	74	30	35

- **Step 5:** Simulating the experiment
  - *Select random numbers from Table 13.4*
  - *The number we select will have a corresponding range in Table 13.3*
  - *Use the daily demand that corresponds to the probability range aligned with the random number*

TABLE 13.5 Ten-Day Simulation of Demand for Radial Tires

DAY	RANDOM NUMBER	SIMULATED DAILY DEMAND
1	52	3
2	37	3
3	82	4
4	69	3
5	98	5
6	96	5
7	33	2
8	50	3
9	88	5
10	90	5

- Note that the average demand from this simulation (3.9 tires) is different from the expected daily demand

$$\text{Expected daily demand}$$

$$= \sum_{i=0}^{5} (\text{Probability of } i \text{ tires}) \times (\text{Demand of } i \text{ tires})$$

$$= (0.05)(0)+(0.10)(1)+(0.20)(2)+(0.30)(3)+(0.20)(4)+(0.15)(5)$$

$$= 2.95 \text{ tires}$$

- In many real-world inventory situations demand and lead time are variables
- Accurate analysis is difficult without simulation
- An inventory problem with two decision variables and two probabilistic components
- The owner of a hardware store wants to establish
  - *Order quantity and reorder point decisions*
  - *Product that has probabilistic daily demand and reorder lead time*

Example: Simkin's Hardware Store

- Find a good, low-cost inventory policy for the Ace electric drill
- Simkin identifies two types of variables
  - Controllable inputs
    - Order quantity
    - Reorder points
  - Uncontrollable inputs
    - Daily demand
    - Variable lead time

TABLE 13.6 Probabilities and Random Number Intervals for Daily Ace Drill Demand

DEMAND FOR ACE DRILL	FREQUENCY (DAYS)	PROBABILITY	CUMULATIVE PROBABILITY	INTERVAL OF RANDOM NUMBERS
0	15	0.05	0.05	01 to 05
1	30	0.10	0.15	06 to 15
2	60	0.20	0.35	16 to 35
3	120	0.40	0.75	36 to 75
4	45	0.15	0.90	76 to 90
5	30	0.10	1.00	91 to 00
Total	300	1.00		

TABLE 13.7 Probabilities and Random Number Intervals for Reorder Lead Time

LEAD TIME (DAYS)	FREQUENCY (ORDERS)	PROBABILITY	CUMULATIVE PROBABILITY	RANDOM NUMBER INTERVAL
1	10	0.20	0.20	01 to 20
2	25	0.50	0.70	21 to 70
3	15	0.30	1.00	71 to 00
Total	50	1.00		

- The third step is to develop a simulation model
  - *A flow diagram is helpful in this process*
- The fourth step in the process is to specify the values of the variables to be tested
  - *The first policy is an order quantity of 10 with a reorder point of 5*
- The fifth step is to actually conduct the simulation
  - *The process is simulated for a 10-day period*

TABLE 13.8 Simkin Hardware's First Inventory Simulation

(1) DAY	(2) UNITS RECEIVED	(3) BEGINNING INVENTORY	(4) RANDOM NUMBER	(5) DEMAND	(6) ENDING INVENTORY	(7) LOST SALES	(8) ORDER	(9) RANDOM NUMBER	(10) LEAD TIME
1	...	10	06	1	9	0	No		
2	0	9	63	3	6	0	No		
3	0	6	57	ORDER QUANTITY = 10 UNITS			REORDER POINT = 5 UNITS	02	1
4	0	3	94	5	0	2	No		
5	10	10	52	3	7	0	No		
6	0	7	69	3	4	0	Yes	33	2
7	0	4	32	2	2	0	No		
8	0	2	30	2	0	0	No		
9	10	10	48	3	7	0	No		
10	0	7	88	4	3	0	Yes	14	1
				Total	41	2			

- The objective is to find a low-cost solution so Simkin must determine the costs
- Equations for average daily ending inventory, average lost sales, and average number of orders placed

Average ending inventory

$$\frac{41 \text{ total units}}{10 \text{ days}} = 4.1 \text{ units per day}$$

Average lost sales

$$\frac{2 \text{ sales lost}}{10 \text{ days}} = 0.2 \text{ unit per day}$$

Average number of orders placed

$$\frac{3 \text{ orders}}{10 \text{ days}} = 0.3 \text{ order per day}$$

Analysing Simkin

- Simkin is open 200 days a year
- Estimated ordering cost is \$10 per order
- Holding cost is \$6 per drill per year
- Lost sales cost \$8

Daily order cost

$$= \text{(Cost of placing one order)} \times \text{(Number of orders placed per day)}$$

$$= \$10 \text{ per order} \times 0.3 \text{ order per day} = \$3$$

Daily holding cost

$$= \text{(Cost of holding one unit for one day)} \times \text{(Average ending inventory)}$$

$$= \$0.03 \text{ per unit per day} \times 4.1 \text{ units per day} = \$0.12$$

Analysing Simkin

- Simkin is open 200 days a year
- Estimated ordering cost is \$10 per order
- Holding cost is \$6 per drill per year
- Lost sales cost \$8

Daily stockout cost

$$= \text{(Cost per lost sale)} \times \text{(Average number of lost sales per day)}$$

$$= \$8 \text{ per lost sale} \times 0.2 \text{ lost sales per day} = \$1.60$$

Total daily inventory cost

$$= \text{Daily order cost} + \text{Daily holding cost} + \text{Daily stockout cost}$$

$$= \$3 + \$0.12 + \$1.60 = \$4.72$$

- For the year, this policy would cost approximately $944
- Simulation should really be extended for many more days
- Even after a larger simulation, the model must be verified and validated to make sure it truly represents the situation on which it is based
- If we are satisfied with the model, additional simulations can be conducted using other values for the variables
- After simulating all reasonable combinations, Simkin would select the policy that results in the lowest total cost

Other simulation examples

- Simulation of a queuing problem
  - "Port of New Orleans"
- Simulation model for a maintenance policy
  - "Three Hills Power Company"

Other Types of Simulation Models

- Operational gaming refers to simulation involving two or more competing players
  - Best examples are military games and business games
  - Allow the testing of management and decision-making skills in hypothetical situations of conflict

- Systems simulation similar to business gaming
  - Allows users to test various managerial policies and decisions to evaluate their effect on the operating environment
  - Models the dynamics of large systems
    - Corporate operating system
    - Urban government
    - Economic systems
  - Allows ”what-if?” questions to test the effects of various

- It is important that a simulation model be checked to see that it is working properly and providing good representation of the real-world situation
- **Verification** involves determining that the computer model is internally consistent and following the logic of the conceptual model
  - *Answers the question "Did we build the model right?"*
- **Validation** compares a simulation model to the real system it represents to make sure it is accurate
  - *Answers the question "Did we build the right model?"*

- Simulation software tools have been developed to make the process easier
  - Arena
  - ProModel
  - SIMUL8
  - ExtendSim
  - Proof 5

- Excel and add-ins can also be used
  - @Risk
  - Crystal Ball
  - RiskSim
  - XLSim

Questions

